PRODUCT USER MANUAL

For Arctic Ocean Sea Ice Reanalyis Product

ARCTIC_MULTIYEAR_PHY_ICE_002_016
Issue: 1.0

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RECORD TABLE

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
ECMWF	European Centre for Medium Range Weather forecast
GLO	Global
HYCOM	Hybrid Coordinates Ocean Model
WAM	WAve Model
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
HINDCAST	An assessment of the past state of the ocean variables made using numerical models with or without data assimilation. Simulations and analyses are hindcasts.
SIMULATION	An hindcast done without data assimilation.
NRT-ANALYSIS:	An hindcast done with data assimilation for a period very close to present. The quality and the number of available observations it's not optimal (less observations and/or less accurate than in a best-analysis).
BEST-ANALYSIS:	An hindcast done with data assimilation for a period close to present. The quality and the number of available observations

	it's the best available (more observations and/or more accurate than in a NRT-analysis).
neXtSIM	NeXt generation Sea Ice Model

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.
Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

Note that in the case of Arctic products, subsetting selected areas or variables will return data interpolated to regular Lat-Lon projection instead of the Polar stereographic projection indicated below, vector variables are rotated accordingly from x- and y- directions to Eastwards and Northwards components. This re-projection is smoothing slightly the results and slightly increases the land mask. Other solutions are being investigated.

INTRODUCTION

a) Summary

This guide describes the data product files from the Arctic Monitoring and Forecasting Centre, what data services are available to access them, and how to use the files and services.

ARCTIC_MULTIYEAR_PHY_ICE_002_016 is the stand-alone sea ice reanalysis product, based on the same 2D stand-alone sea ice system as the PHY_ICE_002_011 NRT forecast and providing the same sea ice variables. It is using the ARCTIC_MULTIYEAR_PHY_002_003 ocean reanalysis as ocean input data.

b) Polar projections

All products are interpolated onto the same polar stereographic output grid, although at different resolutions. HYCOM uses a staggered Arakawa C grid but all variables are defined at the center of the output grid cell after interpolation.
The same projection is used for all Arctic MFC grids on a spherical Earth, following this definition under the proj4 projection library: (Beware, this is not the WGS84 ellipsoid reference).

projection_stereo:proj4 = "+units=m +proj=stere +a=6378273.0 +b=6378273.0 +lon_0=-45.0 +lat_0=90.0 +lat_ts=90.0 +ellps=sphere" ;

This polar stereographic projection also defines the reference direction of vectors (current and sea ice velocity), which should therefore be rotated if the data is reprojected to a Lat-Long or any other grid.
In ArcGIS and QGIS softwares, the x- and y- coordinates units should be converted from 100 km to meters.

A Python code can rotate ice drift vectors to regular Eastward and Northward velocities

```
import numpy as np
import cartopy.crs as ccrs
import cartopy.feature as cfeature
from cartopy.crs import Projection
from netCDF4 import Dataset


# load data (grid and sea ice velocities)
with Dataset("202301_mm-nersc-MODEL-nextsimf-ARC-RAN-fv00.0.nc") as ds:
   vxsi = ds["vxsi"][0]
   vysi = ds["vysi"][0]
   x = ds["x"][:]
   y = ds["y"][:]
   lon = ds["longitude"][:]
   lat = ds["latitude"][:]
spd = np.hypot(vxsi, vysi)
```

```
x2d, y2d = np.meshgrid(x, y)

# Define projections
proj_stere = ccrs.Stereographic(central_latitude=90,
   true_scale_latitude=90,
                              central_longitude=-45,
   globe=ccrs.Globe(semimajor_axis=6378273, semiminor_axis=6378273))
proj_geo = ccrs.PlateCarree()  # lon-lat

# transform velocities from stereographic projection to lon-lat
usi, vsi  = proj_geo.transform_vectors(proj_stere, x2d, y2d, vxsi, vysi)

# transform velocities back from lon-lat to stereographic projection
vxsi_, vysi_  = proj_stere.transform_vectors(proj_geo, lon, lat, usi, vsi)
```

DESCRIPTION OF THE PRODUCT SPECIFICATION

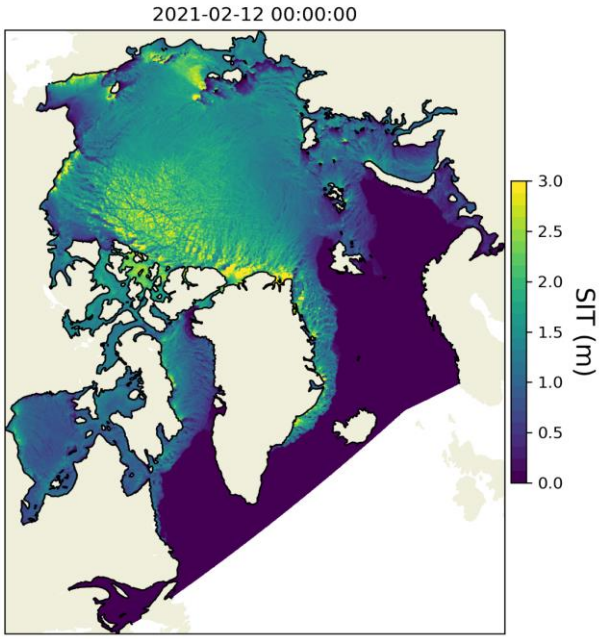
a) General Information

Product Line	ARCTIC_MULTIYEAR_PHY_ICE_002_016
Geographical coverage	North of 63°N and ice-covered North Atlantic Ocean.
Variables	ARCTIC_MULTIYEAR_PHY_ICE_002_016 • Sea Ice (total concentration, thickness, concentration of multi-year ice, concentration of young ice, age, ridged ice fraction and velocity) • Snow depth on sea ice, sea ice albedo
Product Type	Reanalysis
Available time series	ARCTIC_MULTIYEAR_PHY_ICE_002_016 From January 1993 to Y-1
Temporal resolution	Daily and Monthly average fields
Update Frequency	ARCTIC_MULTIYEAR_PHY_ICE_002_016 updated yearly, in November
Horizontal resolution	3 km
Number of vertical levels	ARCTIC_MULTIYEAR_PHY_ICE_002_016: 2D Sea ice only
Format	NetCDF4

b) Production System Description

ARCTIC (180°W-180°E; 63N – 90°N): polar stereographic grid, 2367x2467 cells

This product is regional with an adaptive Lagrangian native triangular mesh covering the Arctic from Bering Strait to the Labrador Sea at 50N (the Pacific side is excluded). It is provided on a standard polar stereographic grid at 3 km resolution. All parameters are interpolated from the native mesh to the same output grid.



Vertical grid	Surface variables only
System Name	neXtSIM-F
Model version	neXtSIM Lagrangian version 2023
Horizontal output Resolution	3 km
Vertical output Resolution	None
Interpolation method	bilinear
Coupling	N/A (stand-alone model)
Sea Ice Model	neXtSIM
Bathymetry	Modified ETOPO 2"
Initial conditions	TOPAZ4 reanalysis plus 2 years of spinup.
Lateral boundary conditions	N/A
Atmospheric forcings	Source: ECMWF ERA5 Frequency: hourly Resolution: 25 km
Ocean forcings	Source: ARCTIC_MULTIYEAR_PHY_002_003 Frequency: daily Resolution: 12.5 km
Tidal constituents	None

Data Assimilation scheme	Nudging
Assimilated observations	SIC (OSI-SAF) Sea ice thickness (CS2SMOS, October to April only)

c) Dataset details

Product Line	ARCTIC_MULTIYEAR_PHY_ICE_002_016	
Geographical coverage	Arctic	
Datasets: cmems_mod_arc_phy_my_nextsim_P1D-m cmems_mod_arc_phy_my_nextsim_P1M-m cmems_mod_arc_phy_my_3km-climatology_P1M-m		
Product Type	Reanalysis	
Update frequency	Yearly	
Ensemble	None	
Available time series	1993- Y-1	
Target delivery time	November	
Temporal resolution (respective datasets)	Daily mean Monthly mean	

Datasets: cmems_mod_arc_phy_my_nextsim_P1D-m cmems_mod_arc_phy_my_nextsim_P1M-m cmems_mod_arc_phy_my_3km-climatology_P1M-m Contain the 2D sea ice variables			
Variable	**Unit**	**Name in file**	**Standard name**
Sea Ice Concentration	1	siconc	sea_ice_area_fraction
Sea Ice Thickness	m	sithick	sea_ice_thickness
Surface snow thickness	m	sisnthick	surface_snow_thickness
Ice drift x-direction velocity	m/s	vxsi	sea_ice_x_velocity
Ice drift y-direction velocity	m/s	vysi	sea_ice_y_velocity
Sea ice albedo	1	sialb	sea_ice_albedo;
Sea ice age	year	siage	sea_ice_age;
Concentration of multi-year ice	1	siconc_my	sea_ice_classification;
Concentration of young ice	1	siconc_young	sea_ice_classification;
Volume fraction of ridged sea ice	1	si_ridge_ratio	sea_ice_volume_fraction_of_ridged_ice;

d) Additional information on some parameters

siage [year]	Sea_ice_age. The volume-averaged age of the sea ice is generally lower than the surface sea ice age observed from satellites.
si_ridge_ratio [1]	Sea_ice_volume_fraction_of_ridged_ice is a statistical indication of the presence of pressure ridges within a given grid cell but will not contain detailed information about individual ridges.

Note that fractions of multi-year ice and young ice share the same CF standard name "sea_ice_classification".

e) Processing information

neXtSIM is a stand-alone sea-ice model that can use winds and currents from a variety of atmospheric and oceanic models (hindcasts or forecasts). Its dynamical core is the Brittle-Bingham-Maxwell (BBM) rheology (Olason et al. 2022). The dynamical equations are solved with a finite element method on a Lagrangian (moving) triangular mesh. The code is a parallelised C++ code (used by Rampal et al., 2019, and presented by Samaké et al., 2017). The domain is a pan-Arctic one, with triangle sides of about 10 km (corresponding to about 7 km square grid resolution).
There is also a thermodynamic component of the code and beneath the ice is a slab ocean with three variables: temperature, salinity and thickness. The temperature and salinity are modified by the heat and salinity fluxes determined by the thermodynamical model as ice melts and freezes and as the model interacts with the atmosphere. The thermodynamical model is a three-category model (detailed in Rampal et al., 2019, Appendix A): open water, newly-formed ice (treated as one ice layer and one snow layer; Semtner, 1976) and older ice (treated as two ice layers and a snow layer; Winton, 2000).
Ocean forcing: To force neXtSIM, we use the ARCTIC_MULTIYEAR_PHY_002_003 (TOPAZ4b) near-surface (30 m) ocean velocity and the mixed layer depth (MLD) directly in the model, while the temperature and salinity in the model's slab ocean are relaxed towards the TOPAZ4b sea surface (3 m) temperature (SST) and salinity (SSS) (respectively) over a time scale of about a month. The thickness of the slab ocean is the MLD from TOPAZ4b.
Atmospheric forcing: We use ECMWF ERA5 reanalysis of the 10-m wind velocity, the 2-m air and dew point temperatures (the latter is used to determine the specific humidity of air for the latent heat flux calculation), the mean sea level pressure, the long- and short-wave downwelling radiation, and the total precipitation (this becomes snow if the temperature is below 0◦C).
Data Assimilation: The assimilation is performed using a data insertion method — an updated variable is calculated as a weighted function of the forecast variable and observations. Sea ice concentrations (OSI-SAF SSMIS) are assimilated on a daily basis. When available (October-April), daily (with a 7-day moving window) sea ice thickness observations from CS2-SMOS are also assimilated.

For more details on the present model setup, see NRT product 002_011.

f) Catalogue updating

N/A.

Yearly updates are provided in November after the update of the MULTIYEAR_PHY_002_003 product.

g) Time averaging

Daily averages are centered at 12:00 UTC.
Monthly averages are centered at the middle of the month.
The reference period of the climatology dataset is 1993-2022.

h) Product use and impact

The TOPAZ5 forecast and TOPAZ4b reanalysis are intended to address the broadest range of users in the Nordic Seas and Arctic.

The **sea ice variables are duplicated** in the TOPAZ and neXtSIM forecast and reanalysis products. The TOPAZ products are using a coupled ice-ocean model, which is of interest for users requiring consistency between ocean and sea ice variables. Users rather interested in more accurate and detailed sea ice variables (in particular sea ice drift) are redirected to the stand-alone neXtSIM sea ice forecast and reanalysis. We intend to close that redundancy by coupling the neXtSIM model to HYCOM in a few years.

FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: "How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center"

The ARC MFC products have been set up through different phases of Copernicus Marine and the file and dataset naming conventions have changed in the meantime. As a result, the tables 14 and 15 below provide a summary of the different files that can be found on the CMEMS server to this date.

Table 15: Multiyear products

Product Name	*Dataset Name*	*File Name*
ARCTIC_MULTIYEAR_PHY_ICE_002_016	cmems_mod_arc_phy_my_nextsim_P1D-m	*20190321_dm-nersc-MODEL-nextsimf-ARC-RAN-fv00.0*
	cmems_mod_arc_phy_my_nextsim_P1M-m	*202201_mm-nersc-MODEL-nextsimf-ARC-RAN-fv00.0.nc*
	cmems_mod_arc_phy_my_3km-climatology_P1M-m	*clim01_neXtSIM3km_reanalysis-cmems.nc*

An exhaustive description of the files nomenclature when downloaded through the web portal is available here: Files Nomenclature and Format

a. FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [GB]
cmems_mod_arc_phy_my_nextsim_P1D-m	*20190321_dm-nersc-MODEL-nextsimf-ARC-RAN-fv00.0*	0.11
cmems_mod_arc_phy_my_nextsim_P1M-m	*202201_mm-nersc-MODEL-nextsimf-ARC-RAN-fv00.0.nc*	0.11
cmems_mod_arc_phy_my_3km-climatology_P1M-m	*clim01_neXtSIM3km_reanalysis-cmems.nc*	0.11

b. STRUCTURE OF FILES

An example of header of output NetCDF files is inserted in **annex,** identical for each dataset.

REFERENCES

Quality Information Documents

https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-ARC-QUID-002-016.pdf

ANNEX

Sample file:

```
netcdf netcdf \20220201_dm-nersc-MODEL-nextsimf-ARC-RAN-fv00.0 {
dimensions:
        time = UNLIMITED ; // (1 currently)
        x = 2467 ;
        y = 2367 ;
        nv = 2 ;
variables:
        inbyte stereographic ;
                stereographic:grid_mapping_name = "polar_stereographic" ;
                stereographic:latitude_of_projection_origin = 90. ;
                stereographic:longitude_of_projection_origin = -45. ;
                stereographic:scale_factor_at_projection_origin = 1. ;
                stereographic:straight_vertical_longitude_from_pole = -45. ;
                stereographic: semi_major_axis = 6378273. ;
                stereographic: semi_minor_axis = 6378273. ;
                stereographic:proj4 = "+proj=stere +lat_0=90 +lat_ts=90 +lon_0=-45 +x_0=0
+y_0=0 +R=6378273 +ellps=sphere +units=m +no_defs" ;
                stereographic:false_easting = 0. ;
                stereographic:false_northing = 0. ;
        double x(x) ;
                x:standard_name = "projection_x_coordinate" ;
                x:units = "m" ;
                x:axis = "X" ;
        double y(y) ;
                y:standard_name = "projection_y_coordinate" ;
                y:units = "m" ;
                y:axis = "Y" ;
        double longitude(y, x) ;
                longitude:standard_name = "longitude" ;
                longitude:long_name = "longitude" ;
                longitude:units = "degrees_east" ;
        double latitude(y, x) ;
                latitude:standard_name = "latitude" ;
                latitude:long_name = "latitude" ;
                latitude:units = "degrees_north" ;
        double time(time) ;
                time:standard_name = "time" ;
                time:long_name = "simulation time" ;
                time:units = "days since 1900-01-01 00:00:00" ;
                time:calendar = "standard" ;
                time:axis = "T" ;
                time:bounds = "time_bnds" ;
        double time_bnds(time,nv) ;
                time_bnds: units = "days since 1900-01-01 00:00:00" ;
        double siconc(y,x) ;
                siconc:standard_name = "sea_ice_area_fraction" ;
                siconc:long_name = "Sea Ice Concentration" ;
                siconc:units = "1" ;
```

```
                siconc:_FillValue = "-100000000376832." ;
                siconc:cell_methods = "time: mean (interval: 1 hours) area: mean" ;
                siconc:grid_mapping = "stereographic" ;
        double sithick(time, y, x) ;
                sithick:_FillValue = -100000000376832. ;
                sithick:standard_name = "sea_ice_thickness" ;
                sithick:long_name = "Sea Ice Thickness" ;
                sithick:units = "m" ;
                sithick:cell_methods = "time: mean (interval: 1 hours) area: mean" ;
                sithick:grid_mapping = "stereographic" ;
        double sisnthick(time, y, x) ;
                sisnthick:_FillValue = -100000000376832. ;
                sisnthick:standard_name = "surface_snow_thickness" ;
                sisnthick:long_name = "Surface Snow Thickness" ;
                sisnthick:units = "m" ;
                sisnthick:cell_methods = "time: mean (interval: 1 hours) area: mean" ;
                sisnthick:grid_mapping = "stereographic" ;
        double vxsi(time, y, x) ;
                vxsi:_FillValue = -100000000376832. ;
                vxsi:standard_name = "sea_ice_x_velocity" ;
                vxsi:long_name = "Sea Ice X Velocity" ;
                vxsi:units = "m s-1" ;
                vxsi:cell_methods = "time: mean (interval: 1 hours) area: mean" ;
                vxsi:grid_mapping = "stereographic" ;
        double vysi(time, y, x) ;
                vysi:_FillValue = -100000000376832. ;
                vysi:standard_name = "sea_ice_y_velocity" ;
                vysi:long_name = "Sea Ice Y Velocity" ;
                vysi:units = "m s-1" ;
                vysi:cell_methods = "time: mean (interval: 1 hours) area: mean" ;
                vysi:grid_mapping = "stereographic" ;
double sialb(time, y, x) ;
        sialb:_FillValue = -100000000376832. ;
        sialb:standard_name = "sea_ice_albedo" ;
        sialb:long_name = "Sea Ice Albedo" ;
        sialb:units = "1" ;
        sialb:cell_methods = "time: mean (interval: 1 hours) area: mean where sea_ice"
;
        sialb:grid_mapping = "stereographic" ;
double si_ridge_ratio(time, y, x) ;
        si_ridge_ratio:_FillValue = -100000000376832. ;
        si_ridge_ratio:standard_name = "sea_ice_volume_fraction_of_ridged_ice" ;
        si_ridge_ratio:long_name = "Sea Ice Volume Fraction of Ridged Ice" ;
        si_ridge_ratio:units = "1" ;
        si_ridge_ratio:cell_methods = "time: mean (interval: 1 hours) area: mean where
sea_ice" ;
        si_ridge_ratio:grid_mapping = "stereographic" ;
double siage(time, y, x) ;
        siage:_FillValue = -100000000376832. ;
        siage:standard_name = "age_of_sea_ice" ;
```

```
        siage:long_name = "Age of Sea Ice" ;
        siage:units = "years" ;
        siage:cell_methods = "time: mean (interval: 1 hours) area: mean where sea_ice"
;
        siage:grid_mapping = "stereographic" ;
double siconc_my(time, y, x) ;
        siconc_my:_FillValue = -100000000376832. ;
        siconc_my:standard_name = "sea_ice_classification" ;
        siconc_my:long_name = "Sea Ice Area Fraction of Multi-year Ice" ;
        siconc_my:units = "1" ;
        siconc_my:cell_methods = "time: mean (interval: 1 hours) area: mean" ;
        siconc_my:grid_mapping = "stereographic" ;
double siconc_young(time, y, x) ;
        siconc_young:_FillValue = -100000000376832. ;
        siconc_young:standard_name = "sea_ice_classification" ;
        siconc_young:long_name = "Sea Ice Area Fraction of Young Ice" ;
        siconc_young:units = "1" ;
        siconc_young:cell_methods = "time: mean (interval: 1 hours) area: mean" ;
        siconc_young:grid_mapping = "stereographic" ;

// global attributes:
:Conventions = "CF-1.6" ;
:institution = "NERSC, Jahnebakken 3, N-5007 Bergen, Norway" ;
:source = "neXtSIM model fields" ;
:references = "https://marine.copernicus.eu/" ;
:email = "nextsimf@nersc.no" ;
:title = "neXtSIM-F sea ice reanalysis, 3 km daily-averaged fields " ;
:field_type = "Files based on file type moorings" ;
:forecast_type = "Reanalysis" ;
:field_date = "2022-02-01" ;
}
```